Highly Available Data Center Fabric

MC-LAG • LACP (IEEE 802.1AX) • BFD • VRRP • Leaf-Spine

> **The project (resume line)**
> Achieved **99.99% data center fabric availability** by designing **MC-LAG** and **LACP (IEEE 802.1AX)** redundancy with **BFD-triggered failover** and **VRRP first-hop gateway high availability** across **leaf-spine** switches.

This document explains the project in plain language. It starts with a one-page summary, then explains every building block, then walks through how everything works step by step (including what happens when things break), and finally explains the architecture diagram piece by piece.

What is inside

1. Overall summary in simple terms
2. Key terms at a glance
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1. Overall Summary (In Simple Terms)

A **data center** is a building full of servers. The servers talk to each other and to the outside world through a network of switches and cables called the **fabric**. If the fabric stops working, every application running on those servers stops working too, even if the servers themselves are perfectly healthy.

The goal of this project was simple to state: **keep the network up essentially all the time.** The target, 99.99% availability, means the fabric can be unavailable for no more than about **52 minutes in an entire year** (roughly 4 minutes per month).

To reach that target, the design removes every single point of failure and makes the network notice and route around problems in well under a second. Four ideas work together:

Building block	What it does, in one line	Everyday analogy
Leaf-spine fabric	Every rack switch (leaf) connects to every core switch (spine), so there are always several equal paths.	A city grid: if one street is closed you just take the next one.
MC-LAG + LACP	Each server is cabled to two different leaf switches, and the two cables are bundled into one logical link. The two switches pretend to be one switch.	A house with two front doors that share one address. If one door jams, you use the other and the mail still arrives.
BFD	Sends tiny “are you alive?” messages many times per second on every link so failures are noticed in under a second.	A heart-rate monitor instead of waiting for someone to notice the patient fainted.
VRRP	Gives servers one shared, virtual gateway address that is answered by whichever switch is healthy.	A company phone number that rings at whichever receptionist is at the desk.

The result in one paragraph

Any single cable, network card, leaf switch, spine switch or gateway can fail and the servers keep talking. Failures are detected in roughly 150 to 900 milliseconds and traffic is moved to a healthy path automatically, usually so quickly that applications only see a brief pause that TCP retransmission hides. Because no human has to react and no part of the network is a single point of failure, the fabric met its **99.99% availability** goal.

> **One sentence version**
> Two of everything, bundled so both are used at once, watched by a sub-second heartbeat, with a gateway address that never disappears.

2. Key Terms at a Glance

Term	Simple meaning
Leaf switch (ToR)	The switch at the top of each server rack. Servers plug into it. ToR means Top-of-Rack.
Spine switch	A core switch that connects all the leaf switches together. Servers never plug into spines.
Link aggregation / LAG	Treating several physical links as one bigger logical link. Also called a bond, port-channel or EtherChannel.
LACP	Link Aggregation Control Protocol. The “handshake” that both ends use to agree which links belong in the bundle. Defined in IEEE 802.1AX (earlier 802.3ad).
MC-LAG	Multi-Chassis Link Aggregation. A LAG whose links land on two different switches that act as one. Vendor names: vPC (Cisco), MLAG (Arista), VLT (Dell), VSX (Aruba), MC-LAG (Juniper).
Peer-link	The direct link between the two MC-LAG switches used to share state and carry backup traffic.
Keepalive	A separate heartbeat path between the two MC-LAG switches that tells them whether the partner is alive.
BFD	Bidirectional Forwarding Detection. A lightweight protocol that checks whether a path is working, many times per second.
Default gateway	The router address a server sends traffic to when the destination is not on its own subnet.
VRRP	Virtual Router Redundancy Protocol. Lets two routers share one virtual gateway IP and MAC address.
VIP	Virtual IP address. The shared gateway address, for example 10.10.10.1.
ECMP	Equal-Cost Multi-Path. Spreading traffic across several equally good routes at the same time.
Failover / Convergence	Failover is automatically switching from a failed part to a healthy one. Convergence is the time the network takes to settle on new paths after a change.

Term	Simple meaning
Split-brain	Two paired switches both think they are in charge because they cannot see each other.
Gray failure	A partial failure: the link light is on but traffic does not actually pass (one direction broken, software hang, bad optic).

3. What 99.99% Availability Really Means

Availability is the percentage of time a service is working. Each extra “nine” cuts the allowed downtime by ten times.

Availability	Downtime per year	Per month	Per week	Per day
99% (two nines)	3.65 days	7.3 hours	1.7 hours	14.4 minutes
99.9% (three nines)	8.76 hours	43.8 minutes	10.1 minutes	1.44 minutes
99.99% (four nines) – this project	**52.6 minutes**	**4.4 minutes**	**1.0 minute**	**8.6 seconds**
99.999% (five nines)	5.3 minutes	26 seconds	6 seconds	0.86 seconds

How redundancy creates the extra nines

Availability is commonly calculated as **uptime ÷ total time**, or equivalently **MTBF ÷ (MTBF + MTTR)**, where MTBF is the average time between failures and MTTR is the average time to recover.

- **Make recovery near-instant (reduce MTTR).** If a failure is detected and bypassed in under a second by BFD, LACP and VRRP, the outage is so short it barely counts. A human-driven fix might take 30+ minutes; an automatic one takes milliseconds.
- **Put components in parallel.** Two independent parts that are each 99.9% available give a combined availability of 1 − (0.001 × 0.001) = 99.9999% in theory, because both must fail at the same moment to cause an outage.
- **Remove single points of failure.** Every server has two links to two switches, every leaf has links to every spine, and the gateway lives on two switches. Nothing in the path is “only one.”

> **Reality check**
>
> Theoretical numbers are higher than real ones because of shared risks such as software bugs, misconfiguration and power. That is why the design also uses consistency checks, staged changes, monitoring and regular failover testing (see Section 6).

4. Each Component Explained in Detail

This section goes through every part of the design, what it is, why it is there, and the small details that make it work. Each part is explained in plain terms first, then with the technical specifics.

4.1 Leaf-Spine Fabric (the overall shape)

Simple idea: instead of a tree where everything funnels through one or two big core switches, every rack switch connects directly to every core switch. This shape is also called a **Clos network**.

The rules of a leaf-spine fabric

- Every leaf connects to every spine. With 2 spines, each leaf has 2 uplinks (or more for extra bandwidth).
- Leaves do not connect to other leaves, except the special MC-LAG peer-link between the two leaves in a pair.
- Spines do not connect to other spines.
- Any server can reach any other server in at most **leaf → spine → leaf**, so delay is low and predictable.

Why this shape helps availability

- There are always several equal paths between any two leaves, one through each spine. Routing uses all of them at once (**ECMP**).
- If a spine fails, you lose only a share of bandwidth (half with 2 spines, a quarter with 4), not connectivity.
- Growth is simple: add more spines for more bandwidth, add more leaves for more racks.

How the leaf-spine links work (the underlay)

- Leaf-to-spine links are **routed (Layer 3)** point-to-point links, not switched Layer 2 links. This avoids Spanning Tree and its blocked links.
- A routing protocol (commonly **eBGP**, or OSPF) tells every switch which subnets live behind which leaf.
- Each switch has a **loopback** address used as a stable identity for routing.
- ECMP spreads flows across uplinks using a hash of the packet's addresses and ports, so one conversation stays on one path (no reordering) while different conversations spread out.

4.2 Spine Switches

Simple idea: the highway interchanges of the data center. They only move traffic between leaves and toward the border routers.

- No servers attach to spines. They run routing only.

- They learn, from every leaf, which server subnets live where, and forward toward the correct leaf.
- They run a BFD session with every leaf on every link so a broken link is removed from routing in under a second.
- They connect up to the core/border routers for traffic leaving the data center (north-south traffic).
- Failure impact: a spine failure removes one set of paths. ECMP simply uses the remaining spine(s).

4.3 Leaf Switches (Top-of-Rack)

Simple idea: the on-ramp for a rack. Servers plug into leaves, and leaves send traffic up to the spines.

- Leaves are deployed in **pairs** (Leaf-1A and Leaf-1B). Each pair is one **MC-LAG domain**.
- Leaves are the boundary between Layer 2 (server VLANs) and Layer 3 (routed fabric). In other words, the leaf pair is the servers' **default gateway**, provided through **VRRP**.
- Each leaf has: server-facing ports (bundled with LACP), a peer-link to its twin, a keepalive connection, and routed uplinks to every spine.
- Both leaves in a pair forward traffic at the same time (active-active), so no capacity sits idle.

4.4 Core / Border Routers

Simple idea: the exit gates to the rest of the world: the campus, the internet, the WAN and other data centers.

- They connect to the spines (or to dedicated border leaves) with routed links.
- They advertise a default route into the fabric and advertise the data center subnets outward.
- They are also built in pairs with BFD on their links, so the exit is not a single point of failure.

4.5 Servers and NIC Bonding

Simple idea: each server has two network cards (NICs). One cable goes to Leaf-A and the other to Leaf-B. The operating system bundles them into one logical interface, usually called **bond0**.

- The bond uses **mode 802.3ad** (also called mode 4 or LACP mode) on Linux, or LACP teaming on Windows and in hypervisors.
- The server has **one IP address and one MAC address** on bond0, no matter how many cables are in the bundle.
- **Transmit hash policy** (for example layer3+4) decides which cable each flow uses, so both cables carry traffic.
- **Link monitoring** (miimon, for example every 100 ms) notices a dead cable at the physical layer almost instantly.
- **LACP rate fast** makes the server and switch exchange LACP messages every second instead of every 30 seconds.

- If one cable, NIC port or leaf dies, the bond keeps working on the remaining link. The IP address does not change, so applications do not notice.

4.6 LACP – Link Aggregation Control Protocol (IEEE 802.1AX)

Simple idea: LACP is the conversation two devices have to agree, “these cables are one bundle, and they are all plugged into the right place.” Without LACP, a mis-cabled or half-broken link could silently drop traffic.

History of the standard

Link aggregation was first standardized as **IEEE 802.3ad** (2000). It was later moved into its own standard, **IEEE 802.1AX** (first published 2008, revised since), which is why both names are used.

The parts of LACP

Part	Simple explanation
Actor and Partner	“Actor” is me, “Partner” is the device on the other end. Each side describes itself and repeats what it heard from the other.
LACPDU	The small LACP message. It is sent to a special reserved multicast address (01-80-C2-00-00-02) that switches never forward further.
System ID	Who I am: a system priority number plus a MAC address. A bundle only forms if all links reach the same partner System ID.
Key	A number that groups ports that are allowed to bundle together (same speed, same settings).
Port priority / port number	Used to choose which links are active if there are more links than allowed.
Active / Passive mode	Active sends LACPDUs on its own. Passive only answers. At least one side must be Active; best practice is Active on both.
Timers (rate)	Fast: one LACPDU per second, partner declared lost after 3 seconds. Slow: one per 30 seconds, lost after 90 seconds. This design uses **fast**.
State flags	Bits in each LACPDU that say: in sync, collecting (receiving), distributing (sending), timed out, defaulted or expired.
Min-links	Optional setting: bring the whole bundle down if fewer than N members are up, so traffic moves elsewhere instead of squeezing through too little bandwidth.

How LACP builds a bundle, step by step

1. Each side sends LACPDUs on every member cable, describing its System ID, key and port.
2. Each side checks that every cable reaches the **same partner System ID** with compatible settings.
3. When both sides agree, they set the **in-sync** flag, then turn on **collecting** (receive) and **distributing** (send) for that cable.
4. The cable joins the bundle and starts carrying traffic. Load is shared by hashing each flow onto one member.
5. LACPDUs keep flowing every second. If a member stops hearing its partner for 3 seconds, or the physical link drops, that member is removed and traffic is re-hashed onto the remaining members.

> **Why LACP instead of a static bundle?**
> A static bundle (mode “on”) never checks the other end. If a cable is plugged into the wrong switch, or one direction of a fiber breaks, traffic is silently lost. LACP detects these mistakes and refuses to use the bad link.

4.7 MC-LAG – Multi-Chassis Link Aggregation

Simple idea: normal LACP bundles cables between **two** devices. But if both server cables go to the **same** switch, that switch is a single point of failure. MC-LAG lets the two cables go to **two different switches** while the server still believes it is talking to **one** switch.

How two switches pretend to be one

- Both leaves in the pair are configured with the same **MC-LAG domain** and advertise the **same LACP System ID** to the server. The server sees one partner, so LACP happily forms one bundle across both switches.
- Each server-facing bundle has the same **MC-LAG ID** (or port-channel number) on both leaves, so the two halves are linked together.
- Both leaves forward traffic at the same time (**active-active**). There are no blocked links and no Spanning Tree loops.

The peer-link

- A high-capacity link (usually two or more cables bundled) directly between Leaf-A and Leaf-B.
- **Control job:** synchronizes state between the twins: learned MAC addresses, ARP/neighbor tables, multicast (IGMP) membership and port status. That way either switch can deliver to any server immediately.
- **Data job (backup only):** carries traffic for servers attached to only one leaf (**orphan ports**), and for a server whose link to the other leaf has failed.

- **Loop-prevention rule:** traffic that arrives over the peer-link is **not** sent out of an MC-LAG bundle that still has a working local member, because the twin has already delivered it. This keeps everything loop-free without Spanning Tree.

The keepalive (heartbeat)

- A small, separate heartbeat (usually over the out-of-band management network) sent about once per second, with a timeout of a few seconds.
- Its only job is to answer one question: **"Is my partner switch still alive?"** It carries no user traffic.
- It must use a **different path** from the peer-link. Otherwise one fault could cut both and cause split-brain.

Roles, consistency checks and split-brain protection

- One switch is elected **primary**, the other **secondary**. The role matters only when they lose contact.
- **Consistency checks** compare important settings on both sides (VLANs, MTU, LACP mode, speed). If they differ, the bundle is held down or a warning is raised, instead of creating a hard-to-find problem.
- **Peer-link down, keepalive up:** the partner is alive but they cannot sync. The **secondary shuts its MC-LAG member ports**, so servers use only the primary. Bundles stay up on one side and nothing loops.
- **Keepalive down, peer-link up:** no traffic impact, the pair keeps syncing over the peer-link; an alarm is raised.
- **Partner really dead (both peer-link and keepalive silent):** the survivor takes over everything and keeps forwarding.
- **Reload / restore delay:** after a reboot, a leaf waits until its routing and peer-link are fully ready before turning its server ports back on, so it never attracts traffic it cannot deliver yet.

4.8 BFD – Bidirectional Forwarding Detection

Simple idea: BFD is a very fast, very lightweight heartbeat between two neighbors. If several heartbeats in a row are missed, BFD immediately tells routing, "this path is dead, stop using it."

Why BFD is needed

- Routing protocols have their own hellos, but default timers are slow: OSPF waits 40 seconds, BGP up to 180 seconds.
- A fully cut cable is noticed quickly at the physical layer, but **gray failures** are not: one direction of a fiber failing, a hung line card, a faulty optic, or a device in between that keeps the light on. BFD catches these because it tests real packet forwarding in both directions.

How BFD works

Detail	Simple explanation
Standards	RFC 5880 (core protocol), RFC 5881 (single-hop over IP), RFC 5883 (multi-hop), RFC 7130 (micro-BFD on each LAG member).
Packets	Tiny UDP control packets. Single-hop uses UDP port 3784, multi-hop uses 4784, echo uses 3785.
Three numbers	**Desired TX interval** (how often I send), **Required RX interval** (how often I can receive), **Detect multiplier** (how many misses = dead).
Detection time	Negotiated interval × multiplier. Example: 300 ms × 3 = **900 ms**. Aggressive example: 50 ms × 3 = **150 ms**.
Session states	AdminDown, Down, Init, Up. Two neighbors move Down → Init → Up with a three-way handshake.
Modes	Asynchronous mode (both sides send regularly, most common) and Echo mode (packets are looped back by the neighbor's forwarding hardware).
Clients	BFD does not reroute by itself. **BGP, OSPF, static routes and VRRP register as clients** and react the moment BFD reports Down.
Hardware offload	On data center switches BFD usually runs in hardware, so fast timers do not overload the CPU.
Dampening	A flapping link can be held down for a while after repeated failures so it does not keep pulling traffic back and forth.

Where BFD runs in this design

- On every **leaf-to-spine** routed link, tied to BGP or OSPF. A failed link is withdrawn from ECMP in under one second.
- On **spine-to-border** links for fast failover of outside traffic.
- Optionally tied to **VRRP**, so the backup gateway takes over in sub-second time instead of the default ~3.6 seconds.
- Optionally as **micro-BFD** on each member of the peer-link bundle to catch gray failures inside the bundle.

4.9 VRRP – Virtual Router Redundancy Protocol

Simple idea: every server is configured with one default gateway address, for example 10.10.10.1. If that gateway lived on just one switch, losing the switch would cut every server off. VRRP makes 10.10.10.1 a **virtual** address shared by both leaves, so it never goes away.

The parts of VRRP

Part	Simple explanation
Standard	VRRP version 3, RFC 5798, works for both IPv4 and IPv6. Older VRRPv2 (RFC 3768) is IPv4 only.
VRID	Virtual Router ID, 1 to 255. Identifies one VRRP group on a VLAN (for example group 10 on VLAN 10).
Virtual IP (VIP)	The shared gateway address that servers use, e.g. 10.10.10.1.
Virtual MAC	A matching shared MAC address: 00-00-5E-00-01-<VRID> for IPv4 (00-00-5E-00-02-<VRID> for IPv6). It moves with the VIP, so servers never need to change their ARP entry.
Real addresses	Each leaf also has its own real IP on the VLAN, e.g. Leaf-1A 10.10.10.2 and Leaf-1B 10.10.10.3.
Priority	1 to 254 (255 is reserved for the device that actually owns the IP). Highest priority becomes **Master**; ties go to the higher real IP.
Master / Backup	The Master answers for the VIP. The Backup listens and is ready to take over.
Advertisements	The Master sends a hello to multicast 224.0.0.18 (IP protocol 112, TTL 255) every advertisement interval, default 1 second.
Master down interval	How long the Backup waits before taking over: 3 × interval + skew time, where skew = (256 − priority) ÷ 256 × interval. About 3.6 seconds with defaults; sub-second when BFD is linked or faster timers are used.
Preemption	If a higher-priority router comes back, it takes the Master role back. A **preempt delay** stops it from doing so until it is fully ready.
Object tracking	The Master watches its own uplinks. If it loses them, it lowers its priority so the other leaf becomes Master.
Gratuitous ARP / unsolicited NA	When a new Master takes over it announces “the VIP and virtual MAC are here now,” so switches update their tables immediately.

VRRP together with MC-LAG (active-active gateway)

In plain VRRP only the Master forwards traffic. In an MC-LAG pair that would mean traffic hashed to the Backup leaf has to cross the peer-link to reach the Master, wasting bandwidth. Most MC-LAG platforms therefore run VRRP in **active-active** mode: **both leaves accept and route packets sent to the virtual**

MAC, while VRRP still elects a Master to answer ARP and to own the address when the pair loses contact. The result is that every server gets a gateway on the very first switch its packet reaches.

4.10 Underlay Routing and ECMP

- Each leaf announces its server subnets (e.g. 10.10.10.0/24) to both spines using eBGP or OSPF. Both leaves in a pair announce the same subnet, so the spines can send to either one.
- Each spine sees equal-cost routes and uses **ECMP** to share load.
- BFD is attached to every routing session. When BFD says Down, the routing protocol instantly removes the routes learned over that link, and ECMP shifts traffic to the remaining paths.
- Because routes are withdrawn rather than timing out, convergence is measured in milliseconds.

4.11 Management Network and Monitoring

- A separate **out-of-band (OOB) management network** carries the MC-LAG keepalive, SSH access, and monitoring traffic. It stays reachable even if the main fabric has a problem.
- **Monitoring** (syslog, SNMP or streaming telemetry) watches: LACP member up/down, BFD session flaps, VRRP state changes, MC-LAG peer and consistency status, interface errors and optic power levels.
- **Alerts** go to the operations team so a silent redundancy loss (for example, one server cable down while traffic still flows on the other) is fixed before a second failure causes an outage.
- **Availability reporting** compares measured downtime against the 99.99% target.

5. Architecture Diagram

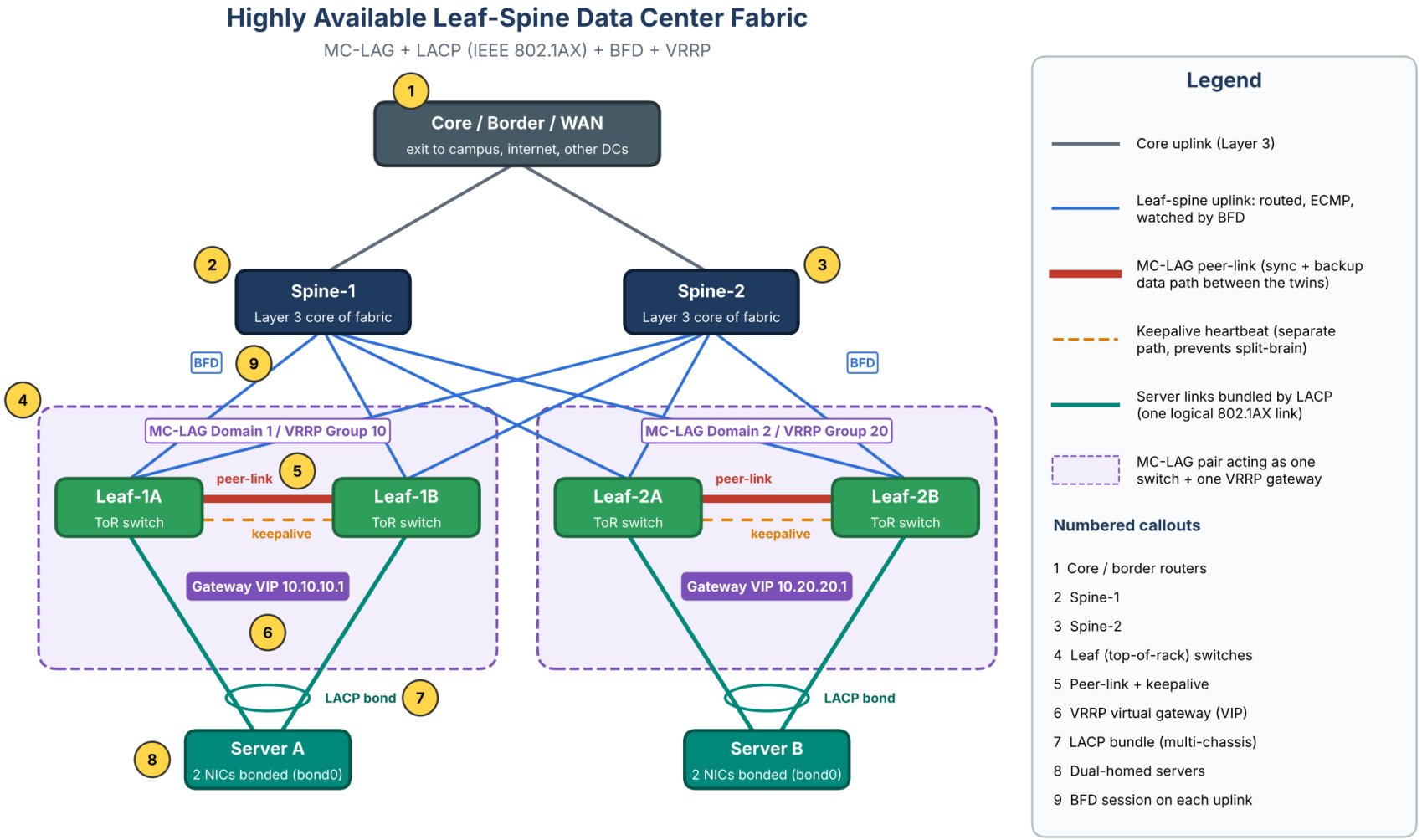


Figure 1. Highly available leaf-spine fabric with MC-LAG, LACP, BFD and VRRP. Numbers match the walk-through on the next pages.

5.1 Walk-Through of the Diagram

Read the diagram from top to bottom: the exit to the outside world at the top, the spines in the middle, the leaf pairs below them and the servers at the bottom. The legend on the right explains every line style. The numbered yellow circles are explained below.

Layer by layer

(1) Core / Border / WAN

The top box is the exit to everything outside the data center. Both spines connect to it with grey routed links. It hands the fabric a default route and announces the data center subnets to the outside. In a real build it is itself a redundant pair.

(2) Spine-1 and (3) Spine-2

The two dark blue boxes are the spines. Each one connects to all four leaves using the blue routed links. No servers connect here. If either spine fails, every leaf still reaches every other leaf through the other spine.

(4) Leaf switches: Leaf-1A, Leaf-1B, Leaf-2A, Leaf-2B

The four green boxes are top-of-rack switches. They come in two pairs. Leaf-1A and Leaf-1B serve the rack containing Server A; Leaf-2A and Leaf-2B serve the rack containing Server B. Each leaf has two uplinks, one to each spine, for a total of eight blue lines.

Purple dashed boxes: MC-LAG domains

Each purple dashed box groups one leaf pair that acts as a single logical switch and a single gateway. The label shows the domain and its VRRP group: Domain 1 with VRRP group 10 (VLAN 10), Domain 2 with VRRP group 20 (VLAN 20).

(5) Peer-link and keepalive

Between the two leaves of each pair are two lines. The **thick red line** is the peer-link that synchronizes MAC/ARP tables and carries backup traffic. The **orange dashed line** is the keepalive heartbeat. It is drawn as a separate line to show it takes a different path (normally the management network), so a single fault cannot cut both.

(6) Gateway VIP

The purple label inside each domain is the shared virtual gateway address. Servers in rack 1 use 10.10.10.1; servers in rack 2 use 10.20.20.1. That address is served by VRRP on both leaves of the pair, so it stays reachable if one leaf fails.

(7) LACP bond (multi-chassis)

The two teal lines from each server go to **different** leaves. The teal ellipse around them shows they are bundled into one logical LACP link. Because both leaves present the same LACP System ID, the server sees a single partner.

(8) Dual-homed servers

Server A and Server B each have two network ports combined into bond0. Each server has one IP address and points to its local VIP as default gateway.

(9) BFD on every uplink

The small BFD tags mark that **every blue leaf-spine link** runs its own BFD session tied to the routing protocol. Two tags are drawn to keep the picture readable, but all eight uplinks have BFD.

What each line style means

Line in the diagram	What it is	Layer	Protected by
Grey solid	Spine to core/border uplink	Layer 3 (routed)	BFD + routing, two spines
Blue solid	Leaf to spine uplink, part of ECMP	Layer 3 (routed)	BFD + ECMP
Thick red	MC-LAG peer-link (usually 2+ cables bundled)	Layer 2 trunk	LACP inside the bundle, keepalive
Orange dashed	MC-LAG keepalive heartbeat	Management network	Separate path from peer-link
Teal solid + ellipse	Server links bundled with LACP across two leaves	Layer 2	LACP fast rate, MC-LAG

Example addressing used in the diagram

Item	Rack 1 (Domain 1)	Rack 2 (Domain 2)
Server VLAN / subnet	VLAN 10 – 10.10.10.0/24	VLAN 20 – 10.20.20.0/24
VRRP group (VRID)	10	20
Virtual gateway (VIP)	10.10.10.1	10.20.20.1
Virtual MAC	00-00-5E-00-01-0A	00-00-5E-00-01-14
Leaf A real IP / priority	10.10.10.2 / 120 (Master)	10.20.20.2 / 120 (Master)

Item	Rack 1 (Domain 1)	Rack 2 (Domain 2)
Leaf B real IP / priority	10.10.10.3 / 100 (Backup)	10.20.20.3 / 100 (Backup)
Example server	Server A – 10.10.10.21	Server B – 10.20.20.21

These values are examples chosen to explain the design. The virtual MAC ends with the VRID in hexadecimal: 10 = 0A, 20 = 14.

Redundancy you can see in the picture

If this fails…	…this takes over	Why it is visible in the diagram
One server cable	The other cable in the same LACP bond	Two teal lines per server
Leaf-1A	Leaf-1B (LACP + VRRP + routing)	Two green boxes per purple domain
Spine-1	Spine-2 via ECMP	Two dark blue spines, each leaf linked to both
One leaf-spine uplink	The leaf's other uplink, triggered by BFD	Two blue lines from every leaf
Peer-link	Keepalive decides; secondary isolates itself	Separate red and orange lines
Gateway on Leaf-1A	VIP moves to Leaf-1B	VIP drawn on the domain, not on one switch

6. End-to-End Workflow

This section follows the life of the system: how it is built, how traffic flows on a normal day, what is constantly running in the background, what happens for every kind of failure, how it recovers and how it is tested.

6.1 Build and Bring-Up (Day 0)

1. **Plan the design.** Decide the number of spines and leaves, link speeds, VLANs and subnets, routing protocol, ASNs and IP addresses.
2. **Cable the fabric.** Every leaf to every spine; a peer-link (two or more cables) between each leaf pair; a management connection from every switch to the out-of-band network; two cables from each server, one to each leaf in its pair.
3. **Configure the underlay.** Give every switch a loopback and point-to-point IPs on the uplinks. Start BGP or OSPF so all loopbacks and subnets are reachable.
4. **Turn on BFD** on every leaf-spine and spine-border link and attach it to the routing sessions. Check that every BFD session is Up.
5. **Form each MC-LAG domain.** Configure the domain ID, the peer-link bundle, the keepalive address over the management network, and the reload/restore delay. Confirm the peer status is Up and all consistency checks pass.
6. **Create the server bundles.** On both leaves, create the same port-channel/MC-LAG ID with LACP in active mode and fast rate, and put it in the server VLAN.
7. **Configure the servers.** Create bond0 in 802.3ad mode with both NICs, fast LACP rate and a layer3+4 hash. Assign the server IP and set the default gateway to the VIP.
8. **Configure VRRP.** On each leaf's VLAN interface, set the real IP, VRRP group, VIP, priority (120 on A, 100 on B), preemption with delay, uplink tracking and optional BFD.
9. **Verify.** LACP shows both members collecting and distributing, MC-LAG shows the bundle active on both peers, VRRP shows one Master and one Backup, ECMP shows two next hops, BFD shows all sessions Up.
10. **Run failover tests** (Section 6.7) before carrying production traffic.

6.2 Normal Day: How a Packet Travels

Example: an application on **Server A (10.10.10.21)** sends data to **Server B (10.20.20.21)**, which sits in a different rack and subnet.

1. **The server decides where to send it.** Server B is on a different subnet, so Server A sends the packet to its default gateway, 10.10.10.1.
2. **ARP for the gateway.** Server A asks "who has 10.10.10.1?" The VRRP Master replies with the **virtual MAC** 00-00-5E-00-01-0A. Server A remembers it.

3. **The bond picks a cable.** Server A's bond hashes the flow (source/destination IP and ports) and chooses one NIC, say the one connected to Leaf-1A. A different flow might use the cable to Leaf-1B.
4. **The leaf routes it.** Leaf-1A receives a frame addressed to the virtual MAC. With active-active VRRP it routes the packet itself, even if it is not the VRRP Master, so no detour across the peer-link is needed.
5. **ECMP picks a spine.** Leaf-1A has two equal routes to 10.20.20.0/24, one via each spine. Its hash picks one, say Spine-2.
6. **The spine forwards it.** Spine-2 knows that both Leaf-2A and Leaf-2B advertise 10.20.20.0/24. It hashes again and picks one, say Leaf-2B.
7. **The destination leaf delivers it.** Leaf-2B already knows Server B's MAC (learned directly or synchronized over the peer-link from Leaf-2A) and sends the frame down its member of Server B's LACP bundle.
8. **Server B receives it** on bond0. The reply follows the same logic in reverse, possibly through different leaves and spine. That is fine; every path is valid.

Traffic to the outside world follows the same path up to a spine, which forwards it to the core/border routers instead of another leaf.

6.3 Background Heartbeats That Always Run

While traffic flows, several health checks run continuously. Together they are the reason failures are caught quickly.

Heartbeat	Between	How often (this design)	Declared failed after
Physical link monitoring	Every cable end	Continuous (hardware)	Milliseconds after light/signal loss
Server bond miimon	Server and its NICs	Every 100 ms	About 100–200 ms
LACPDUs (fast rate)	Server ↔ leaves; across bundles	Every 1 s	3 s of silence
BFD	Leaf ↔ spine, spine ↔ border	Every 300 ms	900 ms (3 misses)
MC-LAG keepalive	Leaf A ↔ Leaf B over management	About every 1 s	A few seconds (platform dependent)
MC-LAG state sync	Leaf A ↔ Leaf B over peer-link	On every change	Peer-link down event

Heartbeat	Between	How often (this design)	Declared failed after
VRRP advertisements	Master → Backup	Every 1 s	About 3.6 s, or sub-second with BFD

6.4 Failure Scenarios, Step by Step

Scenario A – A server cable or NIC port fails

1. The cable between Server A and Leaf-1A breaks.
2. Both the server NIC and Leaf-1A see the physical link go down within milliseconds (LACP fast-rate timeout catches cases where the light stays on, within 3 seconds).
3. The server's bond removes that member and re-hashes all flows onto the cable to Leaf-1B.
4. Leaf-1A marks its half of the MC-LAG bundle as down and tells Leaf-1B over the peer-link.
5. Return traffic arriving at Leaf-1A for Server A is now sent across the peer-link to Leaf-1B (this is one of the peer-link's backup jobs) and delivered.
6. Monitoring raises an alert: "redundancy lost on Server A."

Impact: a few milliseconds of loss for flows on the broken cable. No IP change, no application impact.

Scenario B – A whole leaf switch fails (for example, power loss on Leaf-1A)

1. Every link on Leaf-1A goes down at once.
2. **Servers:** every server bond in rack 1 drops its Leaf-1A member and uses Leaf-1B only.
3. **Spines:** BFD sessions to Leaf-1A go down (or the links drop). BGP/OSPF withdraws routes via Leaf-1A. Spines now send rack-1 traffic only to Leaf-1B.
4. **MC-LAG:** Leaf-1B sees both the peer-link and the keepalive go silent, concludes its partner is dead, and keeps all bundles up as the sole member.
5. **VRRP:** if Leaf-1A was the Master, Leaf-1B stops hearing advertisements and becomes Master (sub-second with BFD, about 3.6 s otherwise). It sends gratuitous ARP for the VIP. With active-active forwarding, Leaf-1B was already routing for the virtual MAC, so traffic hashed to it never stopped.
6. The rack now runs at half its usual bandwidth but with full connectivity.

Impact: sub-second interruption for flows that were on Leaf-1A. Servers keep the same gateway address.

Scenario C – A leaf-to-spine uplink has a gray failure (BFD-triggered failover)

1. The fiber between Leaf-2A and Spine-1 develops a fault: the link light stays on, but packets in one direction are lost.

2. Physical monitoring sees nothing wrong. The BGP session would wait up to its hold time before noticing.
3. **BFD notices** after 3 missed packets at a 300 ms interval: **900 ms**.
4. BFD tells BGP the neighbor is down. BGP immediately withdraws every route learned over that link on both ends.
5. Leaf-2A's ECMP set shrinks from two next hops (Spine-1 and Spine-2) to one (Spine-2). Spine-1 stops sending rack-2 traffic to Leaf-2A and uses Leaf-2B instead.
6. Traffic flows on healthy paths. The faulty link is flagged for repair. BFD dampening keeps it out of service if it starts flapping.

Impact: under one second of loss for the flows hashed to that one link, instead of up to 3 minutes of black-holed traffic without BFD.

Scenario D – A spine switch fails

1. Spine-1 loses power.
2. All four leaves detect the loss of their Spine-1 uplink (link down or BFD down) and remove Spine-1 from ECMP.
3. All traffic between racks and to the core now flows through Spine-2.

Impact: sub-second for flows on Spine-1; total fabric bandwidth is halved (with 2 spines) until it is repaired. Using 4 spines reduces the loss to one quarter.

Scenario E – The MC-LAG peer-link fails (split-brain prevention)

1. Both cables of the peer-link between Leaf-1A and Leaf-1B fail.
2. The keepalive, running over the separate management network, still says "my partner is alive."
3. Both leaves now know the partner is alive but they can no longer synchronize. This is the classic split-brain risk.
4. To stay safe, the **secondary** leaf (say Leaf-1B) **shuts down its MC-LAG server-facing ports** and may withdraw its routes.
5. Each server bond sees its Leaf-1B member go down and uses Leaf-1A only. VRRP and routing settle on Leaf-1A.
6. Operations is alerted to repair the peer-link. When it returns, Leaf-1B resyncs and then re-enables its ports after the restore delay.

Impact: a brief interruption for flows that were on Leaf-1B; no loops, no duplicate traffic, no conflicting gateways.

Scenario F – The VRRP Master loses its uplinks (object tracking)

1. Both of Leaf-1A's uplinks to the spines fail, but Leaf-1A itself stays up.
2. BFD reports both uplinks down. VRRP **tracking** sees this and lowers Leaf-1A's priority (for example from 120 to 90).

3. Leaf-1B (priority 100) is now higher and, with preemption, becomes Master and announces the VIP.
4. Traffic arriving at Leaf-1A from servers can still reach the fabric by crossing the peer-link to Leaf-1B, and many platforms also bring down the MC-LAG ports of a leaf with no uplinks so servers stop sending to it at all.

Impact: sub-second. Without tracking, Leaf-1A would remain Master while being unable to forward to the fabric.

Scenario G – Only the keepalive fails

The management network path breaks, but the peer-link is fine. The leaves still synchronize over the peer-link, so **nothing changes for traffic**. An alarm is raised, because if the peer-link failed next, the pair could not safely tell “peer-link down” from “partner dead.”

6.5 Failure Summary

Failure	Detected by	Typical detection	Who takes over	User impact
Server cable / NIC	Link loss, LACP	ms (up to 3 s)	Other bond member	None to minimal
Whole leaf	Link loss, BFD, keepalive, VRRP	< 1 s	Twin leaf	Sub-second
Uplink gray failure	**BFD**	900 ms	Other uplink via ECMP	Sub-second
Spine	Link loss, BFD	< 1 s	Other spine(s)	Sub-second, less bandwidth
Peer-link	MC-LAG + keepalive	~1–3 s	Primary leaf only	Brief, no loops
VRRP Master uplinks	BFD + tracking	< 1 s	Backup becomes Master	Sub-second
Keepalive only	MC-LAG	Seconds	Nobody needed	None (alarm)

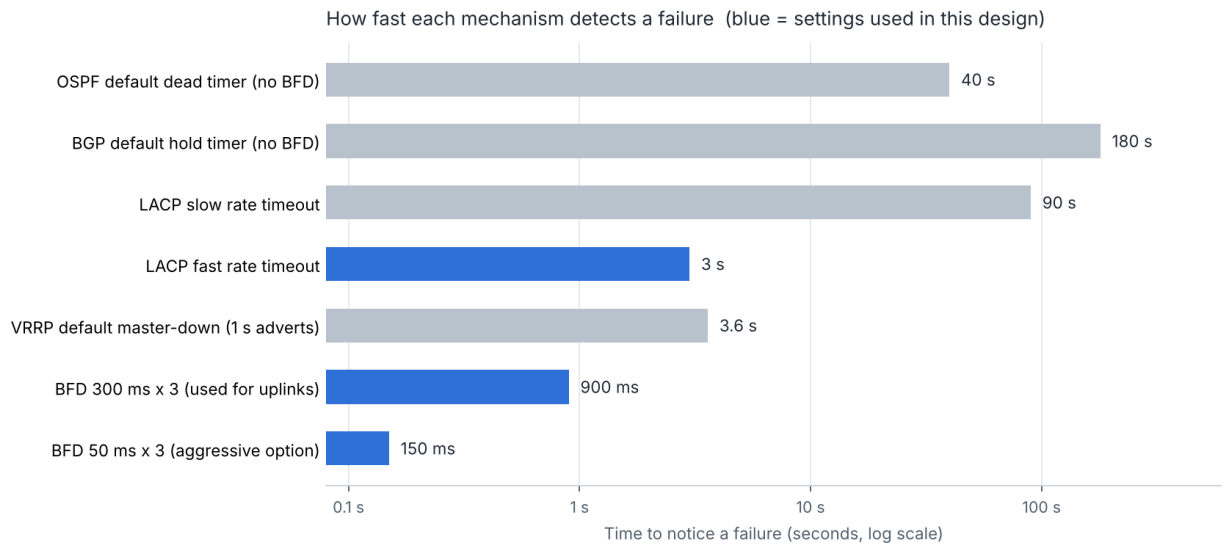


Figure 2. Failure detection time with default timers versus the settings used in this design (log scale).

6.6 Recovery and Failback

Coming back is just as important as failing over. A device that rejoins too early can attract traffic before it is ready and cause a second outage.

- **Restore / reload delay (MC-LAG):** a rebooted leaf keeps its server ports down until routing and the peer-link are fully up and synchronized (for example 3–5 minutes).
- **Preempt delay (VRRP):** the higher-priority leaf waits (for example 60 seconds) before taking the Master role back.
- **BFD and link dampening:** a link that flaps repeatedly is held down for a while so it does not keep pulling traffic back and forth.
- **Routing:** once BFD and the routing session are Up again, routes are re-advertised and ECMP automatically uses the restored path.
- **LACP:** the repaired cable goes through the normal LACP handshake and rejoins the bundle only when both ends agree.

6.7 Testing and Proving 99.99%

1. **Baseline:** run continuous traffic (for example a ping every 10 ms and an iperf stream) between servers in different racks and record packet loss.
2. **Pull one server cable** and measure loss (Scenario A). Reconnect and confirm it rejoins the bundle.
3. **Power off one leaf** and measure loss (Scenario B). Power it back on and confirm the restore delay prevents a second blip.
4. **Shut or break one uplink** and confirm BFD detects it in under one second (Scenario C).
5. **Reload a spine** (Scenario D).

6. **Disconnect the peer-link** and confirm the secondary isolates itself with no loop (Scenario E).
7. **Disconnect the keepalive** and confirm there is no traffic impact (Scenario G).
8. **Record the results** in a table of detection time and packet loss per scenario. Repeat after every software upgrade.
9. **Track real incidents** over the year and calculate availability as uptime ÷ total time to confirm the 99.99% target.

7. Example Configuration

The snippets below show what the important settings look like. They are written in an Arista EOS-style syntax for illustration; command names differ between vendors, so always check your platform's documentation.

Leaf-1A: MC-LAG domain, peer-link and keepalive

```
! ---- MC-LAG peer-link and keepalive (Leaf-1A) ----
vlan 4094
   trunk group MLAGPEER
interface Port-Channel10
   description MC-LAG peer-link to Leaf-1B
   switchport mode trunk
   switchport trunk group MLAGPEER
interface Vlan4094
   ip address 192.168.255.1/30
mlag configuration
   domain-id DOMAIN1
   local-interface Vlan4094
   peer-address 192.168.255.2
   peer-address heartbeat 172.16.0.12 vrf MGMT
   peer-link Port-Channel10
   reload-delay mlag 300
```

Leaf-1A and Leaf-1B: multi-chassis LACP bundle to Server A

```
! ---- Server A bundle (same on Leaf-1A and Leaf-1B) ----
interface Ethernet1
   description Server-A NIC
   channel-group 11 mode active
   lacp timer fast
interface Port-Channel11
   switchport access vlan 10
   mlag 11
```

Leaf-1A: VRRP group 10 with tracking

```
! ---- VRRP gateway (Leaf-1A; Leaf-1B uses .3 and priority 100) ----
interface Vlan10
   ip address 10.10.10.2/24
   vrrp 10 priority-level 120
   vrrp 10 ipv4 10.10.10.1
   vrrp 10 advertisement interval 1
   vrrp 10 preempt delay minimum 60
   vrrp 10 tracked-object UPLINKS decrement 30
```

Leaf-1A: BFD tied to BGP with ECMP

```
! ---- BFD on routed uplinks ----
interface Ethernet49
   description Uplink to Spine-1
   no switchport
   ip address 10.0.1.1/31
   bfd interval 300 min-rx 300 multiplier 3
router bgp 65101
   neighbor 10.0.1.0 remote-as 65000
   neighbor 10.0.1.0 bfd
   maximum-paths 4 ecmp 4
```

Server A: Linux LACP bond

```
# /etc/network/interfaces (Debian/Ubuntu ifupdown)
auto bond0
iface bond0 inet static
    address 10.10.10.21/24
    gateway 10.10.10.1
    bond-slaves eno1 eno2
    bond-mode 802.3ad
    bond-miimon 100
    bond-lacp-rate fast
    bond-xmit-hash-policy layer3+4
```

Useful verification commands

```
show mlag                       ! peer status, consistency
show port-channel 11 detail     ! both members active
show lacp neighbor              ! partner System ID matches
show vrrp brief                 ! Master / Backup
show bfd peers                  ! every session Up
show ip route 10.20.20.0/24     ! two ECMP next hops
cat /proc/net/bonding/bond0     # on the server
```

8. Results and How to Explain This Project

Outcomes

- **99.99% fabric availability** (52.6 minutes or less of downtime per year).
- **No single point of failure** from server NIC to the data center exit.
- **Sub-second failover** for links, leaves, spines and gateways using BFD (around 900 ms, configurable down to about 150 ms) instead of 40–180 seconds with default routing timers.
- **All links active.** MC-LAG and ECMP use every cable instead of leaving backup links idle under Spanning Tree.
- **Zero server reconfiguration on failure.** Servers keep the same IP, gateway VIP and virtual MAC.
- **Safe maintenance.** A leaf or spine can be upgraded or rebooted during business hours with only a brief blip.

Short interview explanation

> "I designed a leaf-spine data center network with no single point of failure. Each server was dual-homed to two leaf switches using LACP, and the two leaves were paired with MC-LAG so the server saw them as one switch. That gave us active-active links and survived a full switch failure. On the routed side, I enabled BFD on every leaf-spine link so a broken or degraded link was removed from routing in under a second instead of tens of seconds. For the server gateway I used VRRP on the leaf pair with uplink tracking, so the default gateway never disappeared. We tested every failure case and reached 99.99% fabric availability."

Likely interview questions

Question	Short answer
Why MC-LAG instead of Spanning Tree?	Spanning Tree blocks redundant links, wasting half the bandwidth, and can take seconds to reconverge. MC-LAG keeps all links active and fails over at LACP/link speed.
What is the peer-link vs the keepalive?	The peer-link syncs state and carries backup traffic. The keepalive is a separate heartbeat that tells whether the partner is alive, preventing split-brain.
Why do you need BFD if links go down anyway?	Clean cuts are fast to detect, but gray failures keep the light on. BFD tests real forwarding in both directions and reacts in under a second.

Question	Short answer
How does VRRP work with MC-LAG?	Both leaves share the VIP and virtual MAC. In active-active mode both route traffic sent to the virtual MAC, so there is no detour over the peer-link.
What happens if the peer-link fails?	If the keepalive says the peer is alive, the secondary shuts its MC-LAG ports so only the primary forwards, avoiding loops and duplicate gateways.
How did you prove 99.99%?	Structured failover tests per component with measured loss, plus tracking real downtime over the year against the 52.6-minute budget.